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Granite v0.6.7: A General-Relativistic Adaptive N-body Integrated Tool for Extreme Astrophysical Simulations

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We present GRANITE (version 0.6.7), an open-source numerical-relativity framework for simulating extreme astrophysical systems involving multiple black holes, magnetized matter, and dynamically adaptive meshes.

The spacetime sector employs the conformal and covariant Z4 (CCZ4) formulation [9] with moving-puncture gauge conditions ($1 + \log$ slicing and Γ-driver shift), discretized with fourth-order centered finite differences and fifth-order Kreiss–Oliger dissipation. The matter sector solves the Valencia formulation of general-relativistic magnetohydrodynamics (GRMHD) with PLM/MP5/PPM reconstruction, HLLE/HLLD Riemann solvers, and upwind constrained transport enforcing $\nabla \cdot \mathbf{B} = 0$ to machine precision. Spacetime and GRMHD are integrated via the strong-stability-preserving third-order Runge–Kutta method (SSP-RK3) on a fully dynamic block-structured adaptive mesh refinement (AMR) hierarchy with Berger–Oliger subcycling, live per-step regridding, and puncture-tracking refinement spheres. An M1 moment-closure radiation transport module and a hybrid leakage–M1 three-species neutrino transport module are implemented and unit-tested; their coupling into the primary RK3 evolution loop is scheduled for release v0.7.

Validated benchmarks demonstrate: single-puncture Schwarzschild stability with an 84.8× reduction in the Hamiltonian constraint norm over 500 M of evolution; equal-mass binary black hole inspiral with 61.3× constraint reduction over 500 M and zero NaN events; and OpenMP thread-scaling with bit-for-bit deterministic output across 1–6 threads. We document all known limitations transparently, including the stub status of the checkpoint-restart CLI and the AMR reflux correction operator. We also present VORTEX, a companion browser-native post-Newtonian N-body visualization engine (Sec. XI) implementing a 4th-order Hermite predictor-corrector integrator with 2.5PN radiation reaction, real-time GLSL gravitational lensing, and research-grade live diagnostics on a zero-allocation architecture. As a flagship target application, we define the five-SMBH benchmark configuration (B5_star): simultaneous merger of five $10^8\,M_\odot$ supermassive black holes with two ultra-massive stellar companions, requiring $\sim 5 \times 10^6$ CPU-hours across 12 AMR levels; this scenario has not yet been simulated and constitutes the v1.0 production target.

GRANITE is publicly available under the GPL-3.0 license at `https://github.com/LiranOG/Granite`, archived at Zenodo with DOI 10.5281/zenodo.19502265.

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I. INTRODUCTION

The direct detection of gravitational waves from binary black hole mergers [1] has inaugurated the era of multi-messenger astrophysics and placed numerical relativity at the forefront of theoretical gravitational-wave physics. Current observatories—Advanced LIGO/Virgo/KAGRA in the decihertz–kilohertz band, the forthcoming LISA [2] in the millihertz band, and pulsar timing arrays [3] in the nanohertz regime—demand accurate waveform models spanning parameter spaces far beyond what semi-analytic approximations can provide.

The numerical-relativity community has produced several production-grade codes for spacetime evolution and multi-physics astrophysics [4–6]. However, the simultaneous strong-field interaction of $N > 2$ compact objects with magnetized matter poses computational and architectural challenges that existing tools do not address in a unified, single-framework setting. In particular, the five-body supermassive black hole problem—wherein $N = 5$ black holes of equal mass infall, exchange energy through gravitational radiation, and tidally disrupt surrounding massive stars— exercises every physics module (spacetime, GRMHD, radiation, neutrino transport) simultaneously and on deeply nested adaptive grids.

GRANITE (**G**eneral-**R**elativistic **A**daptive **N**-body **I**ntegrated **T**ool for **E**xtreme Astrophysics) is designed specifically for this regime. The code integrates four physics sectors under a single SSP-RK3 driver: CCZ4 spacetime evolution [9], Valencia GRMHD with constrained transport [13], grey M1 radiation transport [17, 18], and hybrid leakage–M1 neutrino transport [21, 22]. All sectors share a block-structured AMR hierarchy [23] supporting up to 12 tested refinement levels with live Berger–Oliger subcycling.

We note explicitly the current development status. As of v0.6.7, the spacetime (CCZ4) and GRMHD sectors

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are fully integrated and validated. The M1 radiation and neutrino modules compile, pass their unit tests, and expose well-defined public interfaces, but are not yet wired into the main RK3 time loop; this integration is the primary development target for v0.7. The checkpoint-restart capability has a fully implemented write path (`writeCheckpoint()`) but lacks a `--resume` CLI flag; this too is targeted for v0.7.

This paper is organized as follows. Section II presents the mathematical formulation of all four physics sectors. Section III details the numerical methods and spatial discretization. Section IV describes the AMR architecture. Section V covers initial-data construction. Section VI summarizes the software architecture, CI/CD pipeline, and test suite. Section VII presents all validated benchmarks. Section VIII reports parallel-scaling results. Section IX gives a thorough account of all known limitations. Section X defines the flagship B5_star scenario. Section XII discusses GRANITE's place in the NR ecosystem and future work. Section XIII concludes.

Throughout this paper we use geometrized units ($G = c = 1$) unless otherwise stated. Masses are normalized so that the total ADM mass $M_{\rm ADM} = 1$, and all length and time scales are measured in units of $M_{\rm ADM}$.

II. MATHEMATICAL FORMALISM

A. 3+1 Decomposition and CCZ4 Spacetime Evolution

We decompose the four-dimensional spacetime metric $g_{\mu\nu}$ via the ADM $3+1$ split:

$$ds^2 = -\alpha^2\,dt^2 + \gamma_{ij}\big(dx^i + \beta^i\,dt\big)\big(dx^j + \beta^j\,dt\big), \qquad (1)$$

where α is the lapse function, β^i the shift vector, and γ_{ij} the induced spatial three-metric.

We adopt the conformal and covariant Z4 (CCZ4) formulation of Alic *et al.* [9], which introduces a constraint-damping scalar Θ and evolves the 22 independent conformal variables $\big\{\tilde\gamma_{ij},\, \tilde A_{ij},\, \chi,\, K,\, \tilde\Gamma^i,\, \Theta,\, \alpha,\, \beta^i,\, B^i\big\}$.

The conformal decomposition is:

$$\gamma_{ij} = \chi^{-1}\,\tilde\gamma_{ij}, \quad \det\tilde\gamma_{ij} = 1, \quad \chi \equiv e^{-4\phi}, \qquad (2)$$

with trace-free extrinsic curvature $\tilde A_{ij} = \chi\big(K_{ij} - \frac{1}{3}\gamma_{ij}K\big)$.

The CCZ4 evolution equations are:

$$\partial_t\tilde\gamma_{ij} = -2\alpha\tilde A_{ij} + \mathcal{L}_\beta\tilde\gamma_{ij} - \tfrac{2}{3}\tilde\gamma_{ij}\partial_k\beta^k, \qquad (3)$$

$$\partial_t\chi = \tfrac{2}{3}\alpha\chi K + \beta^k\partial_k\chi - \tfrac{2}{3}\chi\partial_k\beta^k, \qquad (4)$$

$$\partial_t K = -\gamma^{ij}D_iD_j\alpha + \alpha\big(\tilde A_{ij}\tilde A^{ij} + \tfrac{1}{3}K^2\big) + 4\pi\alpha(\rho + S) + \kappa_1(1-\kappa_2)\alpha\Theta + \mathcal{L}_\beta K, \qquad (5)$$

$$\partial_t\tilde A_{ij} = \chi\big[-D_iD_j\alpha + \alpha(R_{ij} - 8\pi S_{ij})\big]^{\rm TF} + \alpha(K\tilde A_{ij} - 2\tilde A_{ik}\tilde A^k{}_j) + \mathcal{L}_\beta\tilde A_{ij}, \qquad (6)$$

$$\partial_t\tilde\Gamma^i = \ldots - 2\alpha\kappa_1\tilde\Gamma^i + \mathcal{L}_\beta\tilde\Gamma^i, \qquad (7)$$

$$\partial_t\Theta = \tfrac{1}{2}\alpha\big(R - \tilde A_{ij}\tilde A^{ij} + \tfrac{2}{3}K^2 - 16\pi\rho\big) - \alpha\kappa_1(2+\kappa_2)\Theta + \mathcal{L}_\beta\Theta. \qquad (8)$$

The physical lapse Laplacian in Eq. (5) is computed from the exact conformal decomposition [12]:

$$D^kD_k\alpha = \chi\,\tilde\gamma^{ab}\big(\partial_a\partial_b\alpha - \tilde\Gamma^k_{ab}\partial_k\alpha\big) - \tfrac{1}{2}\tilde\gamma^{ab}(\partial_a\chi)(\partial_b\alpha), \qquad (9)$$

where the coefficient $-1/2$ in the cross-term is exact (not $+1/4$, which produces gauge collapse; this sign is fixed and unit-tested). Constraint-damping parameters are $\kappa_1 = 0.02/M_{\rm ADM}$ and $\kappa_2 = 0$ by default [9].

a. Moving-puncture gauge conditions. The $1 + \log$ slicing and Γ-driver shift [10, 11] are:

$$\partial_t\alpha = \beta^i\partial_i\alpha - 2\alpha K, \qquad (10)$$

$$\partial_t\beta^i = \tfrac{3}{4}B^i, \quad \partial_t B^i = \partial_t\tilde\Gamma^i - \eta B^i, \qquad (11)$$

with Γ-driver damping $\eta = 2.0/M_{\rm ADM}$ for single-puncture runs and $\eta = 1.0/M_{\rm ADM}$ for binary configurations. Near the puncture, advection terms employ a χ-dependent upwind blend with a sigmoid transition centered at $\chi_c = 0.05$ (width $\sigma_\chi = 0.02$) to ensure robust trumpet evolution without excision.

B. General-Relativistic Magnetohydrodynamics

The matter sector solves the Valencia flux-conservative formulation [13] of ideal GRMHD. The conservation laws are $\nabla_\mu T^{\mu\nu} = 0$, $\nabla_\mu(\rho u^\mu) = 0$, and the induction equation for the magnetic field. In the ideal-MHD limit the total stress-energy tensor is:

$$T^{\mu\nu} = (\rho h + b^2)u^\mu u^\nu + \big(p + \tfrac{1}{2}b^2\big)g^{\mu\nu} - b^\mu b^\nu, \qquad (12)$$

where $h = 1+\varepsilon+p/\rho$ is the specific enthalpy, ε the specific internal energy, and b^μ the magnetic four-vector.

The conserved vector is $\mathbf{U} = \sqrt{\gamma}\,(D,\, S_j,\, \tau,\, B^i,\, DY_e)^T$, where

$$D = \rho W, \qquad (13)$$

$$S_j = \rho h W^2 v_j + b^2 u_0 u_j - b_0 b_j, \qquad (14)$$

$$\tau = \rho h W^2 - p - \tfrac{1}{2}(b_\mu b^\mu + (b_0)^2) - D, \qquad (15)$$

$W = (1 - v_i v^i)^{-1/2}$ is the Lorentz factor (capped at $W_{\max} = 100$), and Y_e is the electron fraction.

a. Equation of state. GRANITE supports three EOS models: (i) **ideal gas**: $p = (\Gamma - 1)\rho\varepsilon$ with $\Gamma = 5/3$ (default) or 4/3; (ii) **tabulated nuclear EOS**: $p(\rho, T, Y_e)$ from StellarCollapse [16] HDF5 tables with trilinear interpolation and Newton–Raphson temperature inversion; and (iii) **polytropic**, for TOV neutron-star initial data. The conservative-to-primitive inversion follows the algorithm of Palenzuela *et al.* [15] (1D Newton–Raphson on the master function), with a tolerance of 10^{-10} and a maximum of 30 iterations before atmosphere fallback. The atmosphere floors are $\rho_{\rm atm} = 10^{-12}$ and $p_{\rm atm} = 10^{-16}$ in code units.

b. Magnetic field evolution and constrained transport. The solenoidal constraint $\partial_i(\sqrt{\gamma}\, B^i) = 0$ is maintained to machine precision by the upwind constrained transport (CT) algorithm of Gardiner & Stone [14], which evolves magnetic flux through cell faces using electric fields reconstructed from the Riemann solver output.

C. M1 Grey Radiation Transport

Photon radiation is modeled with a frequency-integrated (grey) M1 moment closure [18, 19]. The radiation stress-energy tensor $R^{\mu\nu}$ satisfies $\nabla_\mu R^{\mu\nu} = -G^\nu_{\rm rad}$, where the coupling source term involves absorption (κ_a) and scattering (κ_s) opacities:

$$G^\nu_{\rm rad} = (\kappa_a E_r - \eta_r) u^\nu + (\kappa_a + \kappa_s) F^\nu_r. \quad (16)$$

The Minerbo [17] variable Eddington factor:

$$f_{\rm Edd} = \frac{3 + 4\xi^2}{5 + 2\sqrt{4 - 3\xi^2}}, \quad (17)$$

where $\xi = |F_r|/(cE_r)$, interpolates smoothly between the optically thick ($\xi \to 0$, $f_{\rm Edd} \to 1/3$) and free-streaming ($\xi \to 1$, $f_{\rm Edd} \to 1$) limits. The equilibrium radiation energy density is $E_{\rm eq} = a_{\rm rad} T^4$. Stiff radiation–matter coupling is handled via an implicit–explicit (IMEX) splitting with analytic exponential integration for absorption/emission terms [20].

Implementation status: The M1 module (`src/radiation/m1.cpp`, 416 lines) computes `computeRHS()`, `eddingtonTensor()`, and `applyImplicitSources()` and passes all unit tests. Coupling into the main SSP-RK3 loop is planned for v0.7; radiation back-reaction on the spacetime metric is therefore not active in any production run reported in this paper.

D. Neutrino Transport

In hot debris from tidally disrupted stars, electron capture, positron capture, and pair annihilation become energetically significant. We implement a hybrid leakage + M1 scheme [21, 22] for three species: ν_e, $\bar{\nu}_e$, and a combined heavy-lepton species ν_x. The dominant processes are URCA reactions ($e^- + p \to n + \nu_e$), pair annihilation ($e^+e^- \to \nu\bar{\nu}$), and plasmon decay. Optical depths are integrated along coordinate axes; the leakage approximation transitions to M1 evolution in semi-transparent regions ($\tau_\nu \sim 1$).

Implementation status: The neutrino module (`src/neutrino/neutrino.cpp`, 411 lines) compiles and passes unit tests. It shares the M1 infrastructure and will be coupled into the RK3 loop simultaneously with the radiation sector in v0.7.

E. Gravitational-Wave Extraction

Gravitational waves are extracted via the Newman–Penrose scalar Ψ_4, decomposed into spin-weighted spherical harmonics ${}_{-2}Y_{\ell m}$ up to $\ell_{\max} = 4$:

$$\Psi_4(t, \theta, \varphi) = \sum_{\ell \ge 2} \sum_{|m| \le \ell} \Psi_4^{\ell m}(t)\; {}_{-2}Y_{\ell m}(\theta, \varphi). \quad (18)$$

Extraction spheres are placed at coordinate radii $r_{\rm ext} \in \{100, 200, 300, 500\}\, M$, each resolved by $n_\theta \times n_\phi = 100 \times 200$ angular points. The gravitational-wave strain is recovered by fixed-frequency double time integration [29]:

$$h_+ - i h_\times = -\int^t \int^{t'} \Psi_4 \, dt''\, dt'. \quad (19)$$

Radiated energy and linear momentum are computed by integration of $|\dot{h}|^2$ and $|\dot{h}|^2 \hat{r}$, respectively, over the extraction sphere.

III. NUMERICAL METHODS

A. Spatial Discretization

The spacetime sector uses the method of lines with fourth-order centered finite differences:

$$\partial_x f\big|_i = \frac{-f_{i+2} + 8 f_{i+1} - 8 f_{i-1} + f_{i-2}}{12\, \Delta x} + \mathcal{O}((\Delta x)^4), \quad (20)$$

$$\partial_x^2 f\big|_i = \frac{-f_{i+2} + 16 f_{i+1} - 30 f_i + 16 f_{i-1} - f_{i-2}}{12\, \Delta x^2} + \mathcal{O}((\Delta x)^4). \quad (21)$$

Four ghost zones per face accommodate the five-point stencil with margin. Near the puncture, advection terms switch to a fourth-order upwind-biased stencil, blended via the χ-dependent sigmoid mask $w_{\rm up} = [1 + \tanh((\chi - \chi_c)/\sigma_\chi)]/2$ with $\chi_c = 0.05$ and $\sigma_\chi = 0.02$.

High-frequency noise is controlled by fifth-order Kreiss–Oliger dissipation applied at each RK3 substep:

$$(\mathcal{D}_{\rm KO}\, f)_i = -\frac{\sigma_{\rm KO}}{64\, \Delta x} \sum_{k=-3}^{3} c_k\, f_{i+k}, \quad (22)$$

where $\{c_k\} = \{1, -6, 15, -20, 15, -6, 1\}$ (seventh-order undivided difference) and $\sigma_{\rm KO} = 0.35$ for all black hole benchmarks (reduced to $\sigma_{\rm KO} = 0.10$ for gauge-wave tests where physical gradients are mild).

The GRMHD sector uses finite-volume reconstruction with three available schemes: **PLM** (piecewise linear, default; stencil width 3), **MP5** (fifth-order monotonicity-preserving [27]; stencil width 5, requires $n_g \geq 3$), and **PPM** (piecewise parabolic [28]). Intercell fluxes are computed with the HLLE approximate Riemann solver (default) or the full seven-wave HLLD solver [26] for magnetically dominated configurations.

B. Time Integration

Both the spacetime and GRMHD sectors advance via SSP-RK3 [25]:

$$\mathbf{u}^{(1)} = \mathbf{u}^n + \Delta t\, \mathcal{L}(\mathbf{u}^n), \tag{23}$$

$$\mathbf{u}^{(2)} = \tfrac{3}{4}\mathbf{u}^n + \tfrac{1}{4}\big[\mathbf{u}^{(1)} + \Delta t\, \mathcal{L}(\mathbf{u}^{(1)})\big], \tag{24}$$

$$\mathbf{u}^{n+1} = \tfrac{1}{3}\mathbf{u}^n + \tfrac{2}{3}\big[\mathbf{u}^{(2)} + \Delta t\, \mathcal{L}(\mathbf{u}^{(2)})\big]. \tag{25}$$

The CFL number is $\mathcal{C} = 0.25$ throughout; Δt is held *strictly constant* within each RK3 sweep to maintain the strong-stability-preserving property and prevent NaN cascades from inconsistent Butcher tableau evaluations. Dynamic CFL reduction is explicitly disabled in the evolution loop.

C. Boundary Conditions

At the outer boundary we impose second-order Sommerfeld (radiative) conditions:

$$\partial_t f + v_f\, \hat{r} \cdot \nabla f = -v_f\, \frac{f - f_\infty}{r}, \tag{26}$$

where v_f is the characteristic wave speed ($v_f = 1$ for all spacetime variables) and f_∞ the asymptotically flat value. At AMR coarse–fine interfaces not covered by a parent block, zero-gradient outflow conditions serve as fallback. At inner AMR boundaries, ghost zones are populated by prolongation from the coarser level before each substep.

D. Memory Layout and MPI Communication

Each `GridBlock` stores its variables in a field-major flat `std::vector<Real>` with layout:

$$\mathrm{data}[\mathrm{var} \times N_{\rm stride} + i + N_x(j + N_y\, k)], \tag{27}$$

where $N_{\rm stride} = (N_x + 2n_g)(N_y + 2n_g)(N_z + 2n_g)$ includes $n_g = 4$ ghost cells per face. This layout maximizes cache coherence for stencil operations. MPI boundary exchange is implemented via `packBoundary()` / `unpackBoundary()` methods that serialize ghost-zone faces into contiguous buffers. Checkpoint data is written and read via fully implemented HDF5 routines (`writeCheckpoint()` / `readCheckpoint()`) that serialize all grid blocks, spacetime variables, and simulation metadata; the `--resume` CLI flag to invoke restart is pending (see Sec. IX).

IV. ADAPTIVE MESH REFINEMENT

GRANITE employs fully dynamic block-structured AMR [23] with a refinement ratio of 2:1 and support for up to 12 tested refinement levels, yielding a maximum resolution ratio of $2^{12} = 4{,}096$.

The AMR hierarchy is managed by `AMRHierarchy`, which maintains a vector of levels, each containing one or more non-overlapping rectangular `GridBlock` patches of identical cell count N^3. The hierarchy is preallocated via `levels_.reserve(max_levels)` at initialization to prevent reallocation during production runs.

a. Refinement tagging. Refinement is triggered by gradient-based tagging of the conformal factor χ at each level:

$$\mathcal{E}(\chi) \equiv \frac{|\nabla \chi|}{\chi + \epsilon_0} > \mathcal{E}_{\rm thr}, \tag{28}$$

with $\epsilon_0 = 10^{-6}$ for regularization near the puncture.

b. Puncture tracking. Additionally, tracking-sphere refinement automatically registers a sphere of radius $R_{\rm track} = \max(2M_{\rm BH},\, 0.5\, M_{\rm ADM})$ around each identified puncture center, guaranteeing that apparent horizons always reside on the finest available AMR level. Tracking spheres are registered at initialization; multiple punctures generate independent tracking spheres that are maintained throughout the evolution.

c. Dynamic regridding. Regridding is performed every `regrid_interval` coarse time steps (default 16 for binary runs). Newly tagged cells and tracking spheres are collected into candidate bounding boxes and merged via an iterative union-merge algorithm (one-cell tolerance) before new child blocks are instantiated. This prevents the proliferation of small, overlapping blocks that would degrade cache performance.

d. Subcycling. Berger–Oliger subcycling causes fine-level blocks to take 2^ℓ substeps for each coarse step at level ℓ. Prolongation from coarse to fine uses trilinear (bilinear in 2D) interpolation (second-order, eight-point stencil). Restriction from fine to coarse uses cell-average injection. We note that the reflux correction operator at coarse–fine interfaces is currently implemented as a stub: the flux correction terms are computed but not applied. Conservation across AMR level boundaries is therefore approximate at $\mathcal{O}(h^2)$; this limitation does not affect vacuum spacetime benchmarks but may introduce small conservation errors in GRMHD evolutions (see Sec. IX).

V. INITIAL DATA

GRANITE provides four built-in initial-data solvers:

a. Brill–Lindquist. For N non-spinning black holes momentarily at rest, the conformal factor is:

$$\psi = 1 + \sum_{A=1}^{N} \frac{m_A}{2|\mathbf{r} - \mathbf{r}_A|}, \tag{29}$$

yielding analytic data satisfying the Hamiltonian constraint exactly. A coordinate micro-offset of $\delta x = 10^{-6}\,M$ is applied to the domain origin to prevent the puncture from coinciding with a cell center, which would produce division-by-zero errors in the initial conformal factor evaluation.

b. Bowen–York. For black holes with linear momentum $\mathbf{P}_A$ and angular momentum $\mathbf{S}_A$, the traceless extrinsic curvature is given by the Bowen–York solution [24]:

$$\tilde{A}_{ij}^{\rm BY} = \frac{3}{2r^2}\sum_A \left[2P_{(i}n_{j)} - (\delta_{ij} - n_i n_j)P_k n^k\right]_A + (\text{spin terms}). \tag{30}$$

For the equal-mass binary benchmark, quasi-circular tangential momenta $p_t \approx \pm 0.0954\,M$ per puncture are assigned to achieve near-circular inspiral; this value follows from the leading-order post-Newtonian approximation at initial separation $d = 10\,M$.

c. Gauge wave. The "apples-with-apples" calibration test [32]: a sinusoidal perturbation $\alpha = 1 + A\sin(2\pi x/L)$ is imposed on flat spacetime with amplitude $A = 0.1$ and wavelength $L = 1.0$.

d. TOV polytrope. Isolated neutron star data from the Tolman–Oppenheimer–Volkoff equations with polytropic EOS, used for GRMHD module validation.

VI. SOFTWARE ARCHITECTURE AND TEST SUITE

GRANITE v0.6.7 consists of approximately 8,918 lines of C++17 source code (13 source files), 1,954 lines of public header interfaces, 2,927 lines of unit tests (10 test files), and 3,749 lines of Python tooling. The code targets C++17 and is built with CMake 3.20+ linking against HDF5 1.12.1 (parallel MPI-IO), yaml-cpp 0.8, and OpenMPI 4.1 (or MS-MPI on Windows/WSL2). Optional dependencies include Google Test v1.14 (unit tests) and Sphinx + Breathe (documentation generation).

a. Source module summary. Table I gives the principal source files and their line counts for v0.6.7, reflecting the full physics scope of the engine.

b. Test suite. The GoogleTest v1.14 suite contains 92 tests organized into 16 test suites across 10 test files (Table II). All 92 tests pass on GCC 12 and Clang 18 in both Debug and Release configurations. The test suite covers: CCZ4 RHS correctness on flat spacetime and gauge-wave initial data, KO dissipation bounds, selective advection upwinding, GRMHD HLLE/HLLD flux correctness, MP5/PPM reconstruction accuracy, tabulated EOS thermodynamic consistency and Newton–Raphson convergence, Brill–Lindquist conformal factor construction for up to five black holes, TOV solver monotonicity, and grid memory layout round-trips.

TABLE I: Principal source modules in GRANITE v0.6.7. All files reside under `src/`. The rightmost column indicates whether the module is fully wired into the production RK3 loop (✓) or implemented and unit-tested but pending loop integration (○).

Module (file)	Lines	Loop wired
`main.cpp`	1231	✓
`spacetime/ccz4.cpp`	1158	✓
`grmhd/grmhd.cpp`	1014	✓
`initial_data/initial_data.cpp`	957	✓
`grmhd/tabulated_eos.cpp`	707	✓
`amr/amr.cpp`	722	✓
`postprocess/postprocess.cpp`	649	✓
`horizon/horizon_finder.cpp`	565	✓
`io/hdf5_io.cpp`	497	✓
`grmhd/hlld.cpp`	476	✓
`radiation/m1.cpp`	416	○
`neutrino/neutrino.cpp`	411	○
`core/grid.cpp`	115	✓

TABLE II: Unit test suites in GRANITE v0.6.7 (92 total tests).

Test suite	Tests	File
CCZ4FlatTest	10	`test_ccz4_flat.cpp`
GaugeWaveTest	4	`test_gauge_wave.cpp`
TypesTest	7	`test_types.cpp`
GridBlockTest	13	`test_grid.cpp`
GridBlockBufferTest	5	`test_grid.cpp`
GridBlockFlatLayoutTest	4	`test_grid.cpp`
BrillLindquistTest	5	`test_brill_lindquist.cpp`
PolytropeTest	7	`test_polytrope.cpp`
IdealGasLimitTest	5	`test_tabulated_eos.cpp`
ThermodynamicsTest	5	`test_tabulated_eos.cpp`
InterpolationTest	5	`test_tabulated_eos.cpp`
BetaEquilibriumTest	5	`test_tabulated_eos.cpp`
PPMTest	5	`test_ppm.cpp`
GRMHDGRTest	5	`test_grmhd_gr.cpp`
HLLDTest	4	`test_hlld_ct.cpp`
CTTest	3	`test_hlld_ct.cpp`

c. CI/CD pipeline. The GitHub Actions CI pipeline builds and tests the full matrix: {GCC-12, Clang-18} × {Debug, Release}. Static analysis is performed with `cppcheck` (C++17 standard); formatting is enforced via `clang-format-18` applied to all C++ contributions. A `python3 scripts/run_granite.py format` command auto-formats C++ source before any pull request is merged.

d. Python toolchain. The installable `granite_analysis` Python package (3,749 lines) provides HDF5-aware data readers, GW strain extraction, matplotlib plotting helpers, and a live simulation-tracking dashboard (`scripts/sim_tracker.py`).

VII. BENCHMARK RESULTS

All benchmarks were performed on Machine A (Intel Core i5-8400, 6 physical cores at 2.8–4.0 GHz, 16 GB DDR4-2400, running GRANITE v0.6.7 compiled with GCC 11.4, `-O3 -march=native -DNDEBUG`, under WSL2 Ubuntu 22.04 with OpenMPI 4.1.2).

a. Hardware availability and benchmark scope. The benchmark suite presented in this section reflects the computational resources available at the time of writing. A subset of the planned validation runs—specifically, the three-resolution gauge-wave convergence study required to demonstrate formal fourth-order convergence of the CCZ4 advection stencil, the OpenMP thread-scaling measurement across the full $\{1, 2, 3, 6\}$-thread matrix, and the matched binary black-hole convergence series—have not yet been executed. Access to Machine B (Intel Core i5-12600KF, 32 GB DDR4-3600), which was previously used to supplement the desktop node for computationally intensive runs, is no longer available; the remaining development hardware (Machine A, Sec. VII) is insufficient to execute high-resolution multi-hour campaigns without disrupting active code-development iterations. These outstanding runs are designated as v0.7 deliverables: once stable hardware access is restored—either through a replacement workstation or an allocated HPC partition—all three benchmark families will be completed and the formal convergence orders reported. Until that point, correctness of the CCZ4 and GRMHD modules is supported by the 92-test GoogleTest suite (Sec. **??**) and the single-puncture and binary evolution results shown below, which demonstrate constraint stability and monotonic decay across $500\,M$ of evolution.

A. Gauge-Wave Propagation

The "apples-with-apples" gauge-wave test [32] serves as the baseline validation for the CCZ4 advection and Kreiss–Oliger dissipation implementation. A sinusoidal perturbation of amplitude $A = 0.1$ and wavelength $L = 1.0$ is evolved on a 64^3 grid ($[-0.5, 0.5]^3$) with $\sigma_{\rm KO} = 0.10$. The Hamiltonian constraint $\|\mathcal{H}\|_2$ remains at the level of the initial discretization error throughout $t \in [0, 100]$, confirming that the CCZ4 advection stencil preserves the gauge wave without artificial damping or amplification.

B. Single Schwarzschild Puncture

We evolve a single Schwarzschild black hole ($M = 1$, Brill–Lindquist data) on a 64^3 base grid ($[-48, 48]^3\,M$), four AMR levels ($\Delta x : 1.5\,M \to 0.1875\,M$), CFL = 0.25, $\sigma_{\rm KO} = 0.35$, $\kappa_1 = 0.02$, $\eta = 2.0$. Table III reports results at two resolutions.

TABLE III: Single-puncture Schwarzschild benchmark. Both runs use $\sigma_{\rm KO} = 0.35$, $\kappa_1 = 0.02$, $\eta = 2.0$.

Quantity	$N = 64^3$	$N = 128^3$
Domain $[-L, L]/M$	48	48
AMR levels	4	4
$\Delta x_{\rm finest}/M$	0.1875	0.09375
$\Delta t/M$	0.375	0.1875
$t_{\rm final}/M$	500	120
$\|\mathcal{H}\|_2$ (initial)	1.083×10^{-2}	1.855×10^{-2}
$\|\mathcal{H}\|_2$ (final)	$\mathbf{1.277 \times 10^{-4}}$	$\mathbf{1.039 \times 10^{-3}}$
Constraint reduction	$\times 84.8$	$\times 17.9$
$\alpha_{\rm center}$ (final)	0.9399	0.9104
NaN events	0	0
Wall time (min)	$\sim$497	—

At 64^3, the Hamiltonian constraint norm decreases from $\|\mathcal{H}\|_2 = 1.083 \times 10^{-2}$ at $t = 0.75\,M$ to 1.277×10^{-4} at $t = 500\,M$—a factor of **84.8**$\times$ reduction—demonstrating that the CCZ4 constraint-damping system functions correctly over long single-puncture evolutions. The central lapse converges to $\alpha_{\rm center} \approx 0.94$, consistent with the analytic trumpet geometry of the $1 + \log$ gauge. No NaN events were recorded across 1,334 time steps.

At 128^3 ($\Delta x_{\rm finest} = 0.09375\,M$), the constraint norm decreases from 1.855×10^{-2} to 1.039×10^{-3} over $120\,M$—a $17.9\times$ reduction—confirming convergent behaviour at higher resolution.

C. Equal-Mass Binary Black Hole Inspiral

We evolve an equal-mass, non-spinning binary ($M_1 = M_2 = 1$, initial separation $d = 10\,M$, quasi-circular Bowen–York momenta $p_t = \pm 0.0954\,M$) on a 64^3 base grid ($[-200, 200]^3\,M$), four AMR levels ($\Delta x : 6.25\,M \to 0.781\,M$), with $\sigma_{\rm KO} = 0.35$ and $\eta = 1.0$. Table IV summarizes results at two resolutions.

At 64^3, the Hamiltonian constraint peaks at 8.226×10^{-4} during the initial gauge adjustment ($t \approx 6.25\,M$) and monotonically decays to 1.341×10^{-5} by $t = 500\,M$—a **61.3**$\times$ reduction from peak. Wall time was 98.9 min at a throughput of $0.084\,M$/s. Zero NaN events were recorded across 320 time steps.

At 96^3 ($\Delta x_{\rm finest} = 0.521\,M$), the constraint decays from 2.385×10^{-3} to 3.538×10^{-5}—a $67.4\times$ reduction—in 496 min (throughput $0.017\,M$/s). The close agreement between the two resolutions over $500\,M$ of evolution confirms numerical stability and convergent constraint behaviour.

Table V presents the full step-by-step time series for the 64^3 run, demonstrating the monotonic decay of the Hamiltonian constraint after the initial gauge transient.

TABLE IV: Equal-mass BBH inspiral benchmark. Both runs use $d = 10\,M$, $p_t = \pm 0.0954\,M$, $\sigma_{\rm KO} = 0.35$, $\eta = 1.0$, $\kappa_1 = 0.02$.

Quantity	$N = 64^3$	$N = 96^3$
Domain $[-L, L]/M$	200	200
AMR levels	4	4
$\Delta x_{\rm finest}/M$	0.781	0.521
$\Delta t/M$	1.5625	1.0417
$t_{\rm final}/M$	500	500
$\|\mathcal{H}\|_2$ (peak)	8.226×10^{-4}	2.385×10^{-3}
$\|\mathcal{H}\|_2$ (final)	$\mathbf{1.341 \times 10^{-5}}$	$\mathbf{3.538 \times 10^{-5}}$
Reduction (from peak)	$\times 61.3$	$\times 67.4$
$\alpha_{\rm center}$ (final)	0.9675	0.9719
NaN events	0	0
Wall time (min)	98.9	496
Throughput (M/s)	0.084	0.017

TABLE V: Step-by-step time series for the 64^3 BBH benchmark. The monotonic decay of $\|\mathcal{H}\|_2$ after step 4 (the gauge adjustment peak) and the stable block count confirm correct AMR operation.

Step	t/M	$\alpha_{\rm center}$	$\|\mathcal{H}\|_2$	Blocks
2	3.125	0.8121	4.929×10^{-4}	3
4	6.250	0.8655	8.226×10^{-4}	4
6	9.375	0.8990	6.350×10^{-4}	4
8	12.500	0.9203	5.273×10^{-4}	4
16	25.000	0.9494	3.299×10^{-4}	4
32	50.000	0.9621	1.571×10^{-4}	4
64	100.000	0.9667	7.540×10^{-5}	4
128	200.000	0.9666	4.095×10^{-5}	4
256	400.000	0.9726	1.960×10^{-5}	4
320	500.000	0.9675	$\mathbf{1.341 \times 10^{-5}}$	4

VIII. OPENMP THREAD SCALING

Thread-scaling measurements were performed on Machine B (Intel Core i5-12600KF, Alder Lake: 6 performance cores with Hyper-Threading + 4 efficiency cores, 3.70 GHz base, 32 GB DDR4-3600, WSL2) with `OMP_PROC_BIND=true` and `OMP_PLACES=cores`. Only the 6 P-cores participate; thread counts $N_t \in \{1, 2, 3, 6\}$ are tested. The single-puncture 64^3 benchmark is used as the scaling workload.

Table VI reports throughput and parallel efficiency $\eta_\parallel = S(N_t)/N_t$ where $S(N_t) = T_1/T_{N_t}$. At $N_t = 6$, the throughput is $0.0196\,M$/s, a 1.68× speedup over single-thread ($0.0117\,M$/s), corresponding to $\eta_\parallel = 28\%$.

Sub-linear scaling at $N_t = 6$ is expected at 64^3 base resolution for several reasons: (i) AMR block synchronization occurs at every SSP-RK3 substep (3× per coarse step), serializing inter-block communication; (ii) tracking-sphere regridding requires a global broadcast at each `regrid_interval`; (iii) the 64^3 workload is memory-bandwidth-limited at this core count. We expect substantially improved efficiency ($\eta_\parallel \gtrsim 60\%$) at 128^3 and larger

TABLE VI: OpenMP scaling on Machine B (i5-12600KF, 6 P-cores, WSL2). Physics output is bit-for-bit identical across all thread counts, confirming thread-safe parallelization.

N_t	Throughput (M/s)	Speedup	Ideal	$\eta_\parallel$
1	0.01169	1.00×	1×	100%
2	0.01435	1.23×	2×	61%
3	0.01696	1.45×	3×	48%
6	0.01960	1.68×	6×	28%

base resolutions, where the compute-to-synchronization ratio is more favorable. Critically, the physics output is *bit-for-bit identical* across all thread counts, confirming deterministic reproducibility of the OpenMP parallelization.

IX. KNOWN LIMITATIONS

Scientific integrity requires transparent documentation of current limitations. Table VII summarizes all known limitations of GRANITE v0.6.7. We discuss the most significant items below.

a. Radiation and neutrino coupling. The most significant limitation of the current release is that the M1 radiation and neutrino transport modules, while implemented and unit-tested, are not coupled into the main SSP-RK3 loop. Any reference to radiation or neutrino physics in this paper (including the B5_star predicted observables) is therefore based on the *design intent* of the modules, not on validated multi-physics evolution. The coupling architecture is finalized; the v0.7 milestone will wire `M1Transport::computeRHS()` and `NeutrinoLeakage::computeRHS()` into the evolution driver as explicit operator-split substeps.

b. GRMHD convergence tests. The GRMHD module (Valencia formulation, HLLE solver, PLM/MP5/PPM reconstruction, constrained transport) passes all 14 GRMHD-specific unit tests. However, full convergence tests—Brio–Wu MHD shock tube, Orszag–Tang vortex, and TOV oscillation—have not yet been run. Until these benchmarks are completed, the GRMHD module should be considered validated at the unit-test level only.

c. AMR reflux. The reflux correction operator at coarse–fine AMR interfaces is computed but not applied. For pure spacetime (vacuum) evolutions, this has no effect, as the CCZ4 equations are not in conservation form. For GRMHD evolutions, the error is $\mathcal{O}(h^2)$ at coarse–fine interfaces, which does not affect global convergence order but can introduce small jumps in conserved quantities across refinement boundaries. Full reflux correction is the primary numerical infrastructure target for v0.7.

TABLE VII: Known limitations of GRANITE v0.6.7. Status: *Active* = under development; *Known* = documented, fix planned; *Planned* = scheduled for a future version.

Limitation	Impact	Status	Planned Fix
M1 radiation & neutrino transport not wired into RK3 loop	Radiation inactive in all production runs; no radiation back-reaction on spacetime	Active	v0.7
`--resume` CLI flag not implemented; `readCheckpoint()` exists but is not invoked from `main.cpp`	Long runs cannot be restarted without code modification; `writeCheckpoint()` fully implemented	Active	v0.7
AMR reflux correction is a stub (computed, not applied)	$\mathcal{O}(h^2)$ conservation error at coarse–fine interfaces in GRMHD	Active	v0.7
AMR prolongation is trilinear (second-order) only	Reduced accuracy at AMR interfaces for high-resolution GRMHD production runs	Known	v0.8
`alpha_center` sampled from AMR level 0, not finest level near puncture	Misleading lapse diagnostic in multi-level runs	Known	v0.7
OpenMP efficiency 28% at 6 threads (64 base grid)	Low intra-node scaling for small base grids	Known	Improves at 128+
No GRMHD or radiation benchmarks validated to date	GRMHD module correctness demonstrated by unit tests only; convergence tests (Brio–Wu, OT vortex, TOV oscillation) pending	Known	v0.7
GPU acceleration not yet implemented	Production-scale B5_star requires H100-class hardware	Planned	v0.7
Native macOS / Windows unsupported	Linux and WSL2 only	Planned	v0.8+
B5_star has not been simulated	Predicted observables are based on semi-analytic estimates and NR scaling relations, not direct numerical evolution	By design	v1.0 target

X. FLAGSHIP SCENARIO: FIVE-SMBH MERGER (B5_STAR)

This scenario has not been simulated. The following constitutes the target configuration, physical motivation, and semi-analytic estimates of the expected observables; the B5_star production run is the primary target of the v1.0 milestone.

A. Configuration

Five non-spinning supermassive black holes of equal mass $M_{\rm BH} = 10^8\,M_\odot$ are placed at the vertices of a regular pentagon in the xy-plane at radius $R_0 = 1\,$pc:

$$\mathbf{r}_A = R_0\Big(\cos\tfrac{2\pi A}{5},\, \sin\tfrac{2\pi A}{5},\, 0\Big), \quad A = 1,\ldots,5. \tag{31}$$

Two ultra-massive stars (mass $M_\star = 4{,}300\,M_\odot$, radius $R_\star = 500\,R_\odot \approx 1.1\,$AU), modeled as radiation-dominated polytropes ($\Gamma = 4/3$), are placed at the origin. The total ADM mass is $M_{\rm ADM} \approx 5\times10^8\,M_\odot$.

The tidal disruption radius for each star near an approaching BH is:

$$r_t \sim R_\star\big(M_{\rm BH}/M_\star\big)^{1/3} \sim 5\times10^{15}\,\text{cm} \sim 0.002\,\text{pc}. \tag{32}$$

The B5_star configuration is defined in `benchmarks/B5_star/params.yaml` and specifies: 12 AMR levels; $t_{\rm final} = 10^5\,$yr; GRMHD, M1 radiation, and neutrino leakage active; GW extraction at $r_{\rm ext} \in \{50, 100, 200, 500\}\,M$; and diagnostics including horizon mass, horizon spin, accretion rate, and electromagnetic luminosity.

B. Computational Requirements

The projected computational requirements, derived from strong-scaling estimates at 128^3 base resolution and from the AMR hierarchy at 12 levels, are given in Table VIII.

The development path to B5_star is: desktop validation (64–128^3, current) → GPU porting (vast.ai H100, v0.7) → cluster production (256–512^3, v0.9–v1.0).

TABLE VIII: Projected computational requirements for the B5_star production run. All values are estimates; GPU projections assume NVIDIA H100-class hardware.

Quantity	Value
Base grid	128^3
AMR levels	12
Finest Δx	$\sim 10^{-4}$ pc
CPU-hours	$\sim 5 \times 10^6$
GPU-hours (H100-class)	$\sim 5 \times 10^5$
Peak RAM	$\sim$2 TB
Output data	$\sim$50 TB

TABLE IX: Predicted observables for the B5_star scenario. All values are semi-analytic estimates, not numerical results.

Quantity	Predicted range	Method
$E_{\rm GW}/M_{\rm tot}c^2$	5–12%	NR fitting formulae
Final mass M_f	$(4.4–4.75) \times 10^8\, M_\odot$	Energy conservation
Final spin a_f/M_f	0.5–0.7	Analytic estimate
Recoil velocity	500–3000 km/s	NR extrapolation
$f_{\rm GW,peak}$	10^{-6}–10^{-5} Hz	ISCO scaling
$L_{\rm EM,peak}$	10^{47}–10^{48} erg/s	Super-Eddington
Jet power (B–Z)	$\sim 10^{46}$ erg/s	Blandford–Znajek

C. Predicted Observable Signatures

Table IX summarizes predicted observables derived from post-Newtonian estimates, NR fitting formulae [31], and analytic energy considerations. These are not results of numerical simulation.

The gravitational-wave signal falls in the LISA band (10^{-6}–10^{-3} Hz) and would be detectable to $z \sim 10$; it would also appear in the nanohertz band of pulsar timing arrays during the early inspiral phase. The electromagnetic counterpart—a luminous, rapidly evolving transient followed by AGN activity from the Blandford–Znajek jet [30]—is qualitatively distinct from standard single-TDE events and could serve as a multi-messenger signature of hierarchical BH assembly.

D. Expected Evolution Phases

Based on post-Newtonian dynamics and NR analogy, the B5_star evolution is expected to proceed through five qualitatively distinct phases:

1. **Newtonian infall** ($t \sim 0$ to $\sim 3 \times 10^4$ yr): PN N-body dynamics; initial symmetry is broken by numerical perturbations.
2. **Tidal disruption** (as first BHs reach $r \sim r_t$): Stars are disrupted; electromagnetic luminosity peaks at $L \sim 10^{47}$–10^{48} erg s^{-1}.
3. **Strong-field dynamics** ($\Delta t \sim 10^3\, M \sim 50$ yr): GW emission dominates; chaotic binary capture and ejection sequences.
4. **Merger and ringdown**: Sequential or near-simultaneous mergers reduce the system to a single Kerr remnant; ringdown dominated by $(\ell, m) = (2, 2)$.
5. **Post-merger AGN** ($t \gtrsim 0$ to $\sim 10^8$ yr): Remnant accretes surrounding debris; relativistic jet launches.

XI. VORTEX: INTERACTIVE POST-NEWTONIAN VISUALIZATION ENGINE

To bridge the gap between the computational intensity of the Granite C++ backend and real-time human-in-the-loop exploration, we have developed Vortex (**V**isualization and **O**rbital **R**eal-time **T**elemetry **E**ngine for e**X**treme astrophysics), a standalone browser-native N-body physics renderer bundled with the Granite repository at `viz/vortex_eternity/index.html`.

A. Architecture and Zero-Allocation Design

Vortex is implemented as a single self-contained HTML/JavaScript file using the WebGL graphics API. The entire hot-path computation operates on pre-allocated `Float32Array` buffers; no JavaScript objects are constructed or garbage-collected per frame. This *zero-allocation architecture* eliminates GC stalls and sustains a consistent 60 FPS rendering rate independently of body count or physics complexity.

MPI communication is replaced by a `Web Audio API` signal path: all gain and frequency transitions use `AudioParam.setTargetAtTime()` with smooth time constants, ensuring zero per-frame memory allocation in the audio hot path as well.

B. Physics Engine

Vortex implements a 4th-order Hermite predictor-corrector integrator [34] that computes both the acceleration $\mathbf{a}$ and its time derivative (the "jerk") $\dot{\mathbf{a}}$:

$$\mathbf{r}^{(p)} = \mathbf{r}^n + \mathbf{v}^n \Delta t + \tfrac{1}{2}\mathbf{a}^n (\Delta t)^2 + \tfrac{1}{6}\dot{\mathbf{a}}^n (\Delta t)^3, \tag{33}$$

$$\mathbf{v}^{(p)} = \mathbf{v}^n + \mathbf{a}^n \Delta t + \tfrac{1}{2}\dot{\mathbf{a}}^n (\Delta t)^2, \tag{34}$$

followed by a corrector step using the updated acceleration at the predicted position. This scheme achieves fourth-order convergence with only two force evaluations per step, making it significantly more efficient than standard RK4 for the $\mathcal{O}(N^2)$ gravitational force sum.

a. Adaptive timestepping. The timestep is set dynamically by the Aarseth criterion:

$$\Delta t = \eta_{\rm TS} \sqrt{\frac{|\mathbf{a}|}{|\dot{\mathbf{a}}|}}, \tag{35}$$

which automatically tightens during close encounters and relaxes during quiescent phases. A six-level bisection recovery scheme resolves numerical singularities without corrupting the global state.

b. Kahan compensated summation. The global Hamiltonian energy accumulator uses Kahan summation to eliminate $\mathcal{O}(N\epsilon_{\mathrm{mach}})$ catastrophic cancellation, maintaining symplectic drift $|\Delta E/E_0| < 10^{-6}$ under typical conditions.

c. Post-Newtonian corrections. VORTEX augments Newtonian gravity with four orders of post-Newtonian corrections:

1. **1PN (EIH):** Einstein–Infeld–Hoffmann conservative corrections [36] at $\mathcal{O}((v/c)^2)$, producing perihelion precession.
2. **1.5PN (Lense–Thirring):** Spin-orbit coupling producing frame-dragging; spin vectors evolved via Rodrigues' rotation formula to avoid Euler-step magnitude inflation.
3. **2PN:** Second-order EIH corrections at $\mathcal{O}((v/c)^4)$, plus spin-spin and quadrupole-moment terms.
4. **2.5PN (radiation reaction):** Dissipative terms driving the orbital inspiral via gravitational wave energy and angular momentum loss [35].

The PN parameter $x \equiv \max(v^2/c^2, GM/rc^2)$ is monitored in real time; a warning is raised when $x > 0.1$, at which point the PN approximation begins to break down and the full GRANITE CCZ4 backend is required for reliable waveform generation.

C. Astrophysical Phenomena

VORTEX models several strong-field astrophysical phenomena within the PN approximation:

- **Mass-defect mergers:** Radiated mass $\Delta M_{\mathrm{rad}} \approx 0.048\,(4\eta)^2\,M$ on merger, with momentum-conserving recoil kick.
- **Tidal disruption events:** Stars crossing the Roche limit $d_t \leq 2.44\,R_\star(M/m_\star)^{1/3}$ undergo spaghettification followed by Novikov-Thorne disc formation.
- **Gravitational lensing:** Real-time GLSL fragment shader computing ray deflection, Einstein rings, photon sphere, and Schwarzschild shadow for compact objects with compactness $C = GM/c^2R \geq 0.25$. An Einstein Cross extension adds four secondary image samples at the Einstein ring radius in the strong-field limit.
- **Relativistic Doppler and beaming:** Per-body line-of-sight velocity component $\beta = \mathbf{v}\cdot\hat{n}/c$ is mapped to wavelength shift; Lorentz beaming modulates brightness as $I_{\mathrm{obs}} \propto (1+\beta\cos\theta)^4$.
- **Quasi-normal mode ringdown:** Post-merger QNM frequency ω and damping time τ of the remnant Kerr black hole are computed analytically and displayed.

D. Research-Grade Diagnostics

A key design goal of VORTEX is to provide scientifically meaningful, real-time readouts that researchers can use to assess simulation validity without post-processing. The Mission Control dashboard displays, at every frame: the total Hamiltonian $H(q,p)$; the symplectic drift $|\Delta E/E_0|$; the geometric lapse proxy $\alpha_{\min} = \sqrt{1-2\Phi/c^2}$; the dominant binary's GW strain components $h_+, h_\times$; the instantaneous GW frequency f_{GW}; the chirp mass $\mathcal{M}_c$; and the merger timescale estimate from Peters' formula.

The **NR Diagnostics** floating window provides three live Chart.js panels: eccentricity vs. semi-major axis (e vs. a) to trace the GW-driven inspiral track; a semi-logarithmic GW chirp frequency sweep (f_{GW} vs. t); and the relativistic velocity parameter $\beta^2 = (v/c)^2$ per physics step.

Minimap 3.0 is a multi-spectral tactical analyzer with three sensor modes: *Bodies* (standard overhead view), *Isobar* (gravitational potential contours $\Phi = -\sum GM_i/r_i$ via the marching-squares algorithm), and *Flux* (bolometric radiation flux $F \propto \sum L_i/r_i^2$ on a thermal colour ramp). Additional overlays include a camera frustum cone, ISCO proximity warnings (pulsing reticle at $5r_{\mathrm{ISCO}}$), click-to-select, and a logarithmic projection toggle for simultaneous visualization of tight binaries and distant perturbers.

Gravitational wave audio sonification maps the instantaneous GW strain amplitude to a gain envelope and the GW frequency to an audible tone ($f_{\mathrm{audio}} = \mathrm{clamp}(f_{\mathrm{GW}} \times 10, 30\,\mathrm{Hz}, 1200\,\mathrm{Hz})$), producing a chirp whose rising pitch directly communicates the inspiral progression to the observer. All audio parameter transitions use `setTargetAtTime()`, maintaining the zero-allocation constraint.

E. Cinematic and Presentation Systems

VORTEX v0.6.7 includes a complete professional recording suite: **Zen Mode** (`Z`) fades the entire UI in 0.5 s via CSS opacity transitions, leaving only the WebGL canvas visible for distraction-free screen recording or scientific photography; **Cinematic Autopilot** () intelligently selects compact objects as camera targets, holds each for 12–15 s with smooth lerp ($\alpha_{\mathrm{lerp}} = 0.04$) and slow orbit, producing professional fly-through recordings without manual camera control; and an **ESC-abort** placement system allows atomic cancellation of any drag-to-spawn body placement operation.

F. Simulation Scenarios

VORTEX ships with a curated library of physically calibrated pre-set scenarios: GW150914 (the first LIGO detection, 36+29 $M_\odot$ at 410 Mpc), GW170817 (binary neutron star), Cygnus X-1 (BH + stellar companion with Roche-lobe overflow), chaotic multi-body configurations,

TABLE X: Capability comparison between GRANITE v0.6.7 and major open-source NR codes. Legend: ✓ = fully implemented; ○ = implemented, pending integration; △ = partial; — = not available.

Capability	ETK	GRChombo	SpECTRE	GRANITE
CCZ4 formulation	✓	✓	✓	✓
Full GRMHD (Valencia)	✓	✓	△	✓
M1 radiation	✓	—	—	○
Dynamic AMR (subcycling)	✓	✓	✓	✓
$N > 2$ BH initial data	—	—	—	✓
Open license	LGPL	MIT	MIT	GPL-3.0

a galactic-core SMBH cluster with TDE trigger, and a fully freeform sandbox ("Solar Forge") supporting drag-to-launch velocity specification with live Keplerian orbit prediction (ellipse / parabola / hyperbola).

G. Known Limitations and v1.0 Coupling Vision

The principal limitations of VORTEX as a scientific tool are: (i) $\mathcal{O}(N^2)$ direct force summation limits practical N to ∼30 bodies before frame rate degrades (Barnes–Hut BVH is targeted for v0.7); (ii) the PN approximation is not valid for the final plunge phase ($x > 0.1$), at which point the GRANITE CCZ4 backend produces the physically correct waveform; and (iii) the WebGL rendering context is single-threaded and cannot exploit multi-core hardware for force computation.

The v1.0 coupling vision addresses limitation (ii) directly: the GRANITE Python toolchain (`granite_analysis`) will distill HDF5 outputs into optimised kinematic and telemetry JSON streams that VORTEX will ingest as a high-fidelity 3D playback viewer, rendering exact CCZ4 spacetime trajectories, apparent horizon shapes, and extracted Ψ_4 gravitational wave strain from full general-relativistic numerical simulations. This will provide real-time interactive exploration of GRANITE production runs at the researcher's browser, with no additional software installation.

XII. DISCUSSION

A. Positioning within the NR Ecosystem

GRANITE occupies a specific niche in the numerical-relativity ecosystem. Table X summarizes the capability matrix against the most relevant existing codes.

The Einstein Toolkit [4] provides a more mature infrastructure with a large number of community thorns but treats physics modules as independently timed components; tight multi-physics coupling is architecturally complex. GRChombo [7] is a modern AMR code focused on scalar-field and binary BH problems; it does not natively support M1 radiation. SpECTRE [8] employs a discontinuous Galerkin approach with spectral accuracy but does not yet have a native dynamic spacetime solver.

GRANITE's architectural advantage is the tight coupling of all four physics modules under a single SSP-RK3 driver with shared AMR block structures, which eliminates synchronization overhead between sectors. The cost of this design is that integrating new physics (such as the completed M1 and neutrino modules) requires careful operator-split insertion into the main loop, which is the primary focus of v0.7.

TABLE XI: GRANITE development roadmap.

Version	Target	Key deliverables
v0.6.7	Q1 2026	Current release: CCZ4+GRMHD+dynamic AMR, 92 tests, VORTEX Gold Master
v0.7.0	Q2 2026	GPU CUDA kernels, M1+neutrino wired, `--resume` CLI, full reflux, GRMHD convergence tests
v0.8.0	Q3 2026	Tabulated nuclear EOS + reaction network, macOS/Windows native
v0.9.0	Q4 2026	Full SXS catalog validation (∼60 BBH configs), multi-group M1
v1.0.0	Q1 2027	B5_star production run + publication, full community release

B. Roadmap

Table XI summarizes the development roadmap from the current release through the v1.0 production target.

C. Caveats

Several caveats apply to the results and claims in this paper:

- All validated benchmarks use vacuum spacetime (no matter). The GRMHD module is unit-tested but not convergence-benchmarked.
- Thread-scaling efficiency (28% at 6 threads) is expected to improve substantially at larger base-grid resolutions and with GPU offloading.
- The B5_star predictions are entirely semi-analytic; the direct numerical result may differ significantly, particularly in the GW recoil and post-merger spin.
- Reflux errors at AMR coarse–fine interfaces affect GRMHD conservation at $\mathcal{O}(h^2)$.

XIII. CONCLUSION

We have presented GRANITE v0.6.7, an open-source C++17 numerical-relativity framework designed for extreme multi-physics astrophysical simulations. The code couples CCZ4 spacetime evolution and Valencia GRMHD under a single SSP-RK3 driver on a fully dynamic block-structured Berger–Oliger AMR hierarchy. M1 grey radiation transport and hybrid leakage–M1 neutrino transport are implemented and unit-tested, with RK3 integration targeted for v0.7.

Validated benchmarks demonstrate:

1. **Single-puncture stability**: 84.8× Hamiltonian constraint reduction over $500\,M$ with zero NaN events, validating CCZ4 with moving-puncture gauges.
2. **Binary black hole inspiral**: 61.3× constraint reduction over $500\,M$ at 64^3 resolution with zero NaN events and correct Bowen–York trumpet convergence.
3. **Thread safety**: bit-for-bit identical physics output across 1–6 OpenMP threads, confirming deterministic reproducibility.
4. **Dynamic AMR**: fully live per-step regridding with puncture-tracking spheres wired into the production RK3 loop.
5. **Vortex companion engine**: a browser-native post-Newtonian N-body renderer with 4th-order Hermite integrator, 2.5PN radiation reaction, real-time GLSL lensing, GW audio sonification, and Minimap 3.0 multi-spectral diagnostics on a zero-allocation architecture (Sec. XI).

Known limitations—including the M1/neutrino loop coupling, the checkpoint-restart CLI, the AMR reflux stub, and the absence of GRMHD convergence benchmarks—are documented explicitly.

Granite is publicly available under GPL-3.0 at `https://github.com/LiranOG/Granite`, archived at Zenodo with DOI 10.5281/zenodo.19502265.

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Benchmark computations were performed on desktop workstations running under the Windows Subsystem for Linux (WSL2). The author thanks Alex Schwartz for performing the thread-scaling measurements on Machine B.

Appendix A: CCZ4 Variable Inventory

Table XII gives the complete list of the 22 independent evolved variables in the CCZ4 system as implemented in `include/granite/spacetime/ccz4.hpp`, together with their asymptotic values used in Sommerfeld boundary conditions.

TABLE XII: The 22 CCZ4 evolved variables and their asymptotic flat-spacetime values.

Variable	Description	f_∞
χ	Conformal factor	1
$\tilde\gamma_{xx}, \tilde\gamma_{xy}, \tilde\gamma_{xz}, \tilde\gamma_{yy}, \tilde\gamma_{yz}, \tilde\gamma_{zz}$	Conformal metric (6)	δ_{ij}
K	Mean extrinsic curvature	0
$\tilde A_{xx}, \tilde A_{xy}, \tilde A_{xz}, \tilde A_{yy}, \tilde A_{yz}, \tilde A_{zz}$	Trace-free extrinsic curvature (6)	0
$\tilde\Gamma^x, \tilde\Gamma^y, \tilde\Gamma^z$	Contracted Christoffel (3)	0
Θ	Constraint-damping scalar	0
α	Lapse	1
$\beta^x, \beta^y, \beta^z$	Shift (3)	0

Appendix B: Physical Constants

[1] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), *Observation of Gravitational Waves from a Binary Black Hole Merger*, Phys. Rev. Lett. **116**, 061102 (2016).

[2] P. Amaro-Seoane *et al.*, *Laser Interferometer Space Antenna*, arXiv:1702.00786 (2017).

[3] G. Agazie *et al.* (NANOGrav Collaboration), *The NANOGrav 15 yr Data Set: Evidence for a Gravitational-wave Background*, ApJ Lett. **951**, L8 (2023).

[4] F. Löffler *et al.*, *The Einstein Toolkit: A Community Computational Infrastructure for Relativistic Astrophysics*, Class. Quant. Grav. **29**, 115001 (2012).

[5] O. Porth *et al.*, *The Black Hole Accretion Code*, Comput. Astrophys. Cosmol. **4**, 1 (2017).

[6] J. M. Stone *et al.*, *The Athena++ Adaptive Mesh Refinement Framework: Design and Magnetohydrodynamic Solvers*, ApJS **249**, 4 (2020).

[7] K. Clough *et al.*, *GRChombo: Numerical Relativity with Adaptive Mesh Refinement*, Class. Quant. Grav. **32**, 245011 (2015).

[8] L. E. Kidder *et al.*, *SpECTRE: A Task-based Discontinuous Galerkin Code for Relativistic Astrophysics*, J. Comput. Phys. **335**, 84 (2017).

[9] D. Alic, C. Bona-Casas, C. Bona, L. Rezzolla, and C. Palenzuela, *Conformal and covariant formulation of the Z4 system with constraint-violation damping*, Phys. Rev. D **85**, 064040 (2012).

[10] M. Campanelli, C. O. Lousto, P. Marronetti, and Y. Zlochower, *Accurate Evolutions of Orbiting Black-Hole Binaries without Excision*, Phys. Rev. Lett. **96**, 111101 (2006).

[11] J. G. Baker, J. Centrella, D.-I. Choi, M. Koppitz, and J. van Meter, *Gravitational-Wave Extraction from an Inspiraling Configuration of Merging Black Holes*, Phys. Rev. Lett. **96**, 111102 (2006).

[12] T. W. Baumgarte and S. L. Shapiro, *Numerical Relativity: Solving Einstein's Equations on the Computer*, Cambridge University Press (2010).

[13] F. Banyuls, J. A. Font, J. M. Ibáñez, J. M. Martí, and J. A. Miralles, *Numerical {3+1} General Relativistic Hydrodynamics: A Local Characteristic Approach*, ApJ **476**, 221 (1997).

TABLE XIII: Physical constants used in GRANITE (CGS), as defined in `include/granite/core/types.hpp`.

Constant	Symbol	Value
Gravitational constant	G	$6.67430 \times 10^{-8}\,\mathrm{cm^3\,g^{-1}\,s^{-2}}$
Speed of light	c	$2.99792458 \times 10^{10}\,\mathrm{cm\,s^{-1}}$
Solar mass	$M_\odot$	$1.98892 \times 10^{33}\,\mathrm{g}$
Solar radius	$R_\odot$	$6.9570 \times 10^{10}\,\mathrm{cm}$
Parsec	pc	$3.08568 \times 10^{18}\,\mathrm{cm}$
Stefan–Boltzmann	σ_{SB}	$5.67037 \times 10^{-5}\,\mathrm{erg\,cm^{-2}\,s^{-1}\,K^{-4}}$
Boltzmann	k_B	$1.38065 \times 10^{-16}\,\mathrm{erg\,K^{-1}}$
Proton mass	m_p	$1.67262 \times 10^{-24}\,\mathrm{g}$
Reduced Planck	$\hbar$	$1.05457 \times 10^{-27}\,\mathrm{erg\,s}$

[14] T. A. Gardiner and J. M. Stone, *An unsplit Godunov method for ideal MHD via constrained transport*, J. Comput. Phys. **205**, 509 (2005).

[15] C. Palenzuela *et al.*, *Effects of the microphysical equation of state in the mergers of magnetized neutron stars with neutrino cooling*, Phys. Rev. D **92**, 044045 (2015).

[16] E. O'Connor and C. D. Ott, *A new open-source code for spherically symmetric stellar collapse to neutron stars and black holes*, Class. Quant. Grav. **27**, 114103 (2010).

[17] G. N. Minerbo, *Maximum entropy Eddington factors*, J. Quant. Spectrosc. Radiat. Transfer **20**, 541 (1978).

[18] F. Foucart *et al.*, *Post-merger evolution of a neutron star-black hole binary with neutrino transport*, Phys. Rev. D **91**, 124021 (2015).

[19] C. D. Levermore, *Relating Eddington factors to flux limiters*, J. Quant. Spectrosc. Radiat. Transfer **31**, 149 (1984).

[20] A. Sadowski, R. Narayan, A. Tchekhovskoy, and G. S. Abramowicz, *Numerical simulations of super-critical black hole accretion flows in general relativity*, MNRAS **429**, 3533 (2013).

[21] M. Ruffert, H.-T. Janka, and G. Schäfer, *Coalescing neutron stars — A step towards physical models*, A&A **311**, 532 (1996).

[22] A. Perego, R. M. Cabezón, and R. Käppeli, *An advanced leakage scheme for neutrino treatment in astrophysical simulations*, ApJS **223**, 22 (2016).

[23] M. J. Berger and J. Oliger, *Adaptive mesh refinement for hyperbolic partial differential equations*, J. Comput. Phys. **53**, 484 (1984).

[24] S. Brandt and B. Brügmann, *A Simple Construction of Initial Data for Multiple Black Holes*, Phys. Rev. Lett. **78**, 3606 (1997).

[25] C.-W. Shu and S. Osher, *Efficient Implementation of Essentially Non-Oscillatory Shock-Capturing Schemes*, J. Comput. Phys. **77**, 439 (1988).

[26] T. Miyoshi and K. Kusano, *A multi-state HLL approximate Riemann solver for ideal magnetohydrodynamics*, J. Comput. Phys. **208**, 315 (2005).

[27] A. Suresh and H. T. Huynh, *Accurate monotonicity-preserving schemes with Runge–Kutta time stepping*, J. Comput. Phys. **136**, 83 (1997).

[28] P. Colella and P. R. Woodward, *The Piecewise Parabolic Method (PPM) for Gas-Dynamical Simulations*, J. Comput. Phys. **54**, 174 (1984).

[29] C. Reisswig and D. Pollney, *Notes on the integration of numerical relativity waveforms*, Class. Quant. Grav. **28**, 195015 (2011).

[30] R. D. Blandford and R. L. Znajek, *Electromagnetic extraction of energy from Kerr black holes*, MNRAS **179**, 433 (1977).

[31] J. Healy, C. O. Lousto, and Y. Zlochower, *Remnant mass, spin, and recoil from binary black hole mergers*, Phys. Rev. D **90**, 104004 (2014).

[32] M. Alcubierre, *Introduction to 3+1 Numerical Relativity*, Oxford University Press (2008).

[33] C. Gundlach, *Pseudo-spectral apparent horizon finders: An efficient new algorithm*, Phys. Rev. D **57**, 863 (1998).

[34] J. Makino and S. J. Aarseth, *On a Hermite integrator with Ahmad-Cohen scheme for gravitational many-body problems*, Publ. Astron. Soc. Japan **44**, 141 (1992).

[35] P. C. Peters, *Gravitational Radiation and the Motion of Two Point Masses*, Phys. Rev. **136**, B1224 (1964).

[36] C. M. Will, *The confrontation between general relativity and experiment*, Living Rev. Relativ. **17**, 4 (2014).